

Online Book Shopping System

Project Overview:

Client requires an Online Book Store to allow users to browse, search, and purchase books online with a user-friendly interface, secure transactions, and account management.

Tasks to complete:

1. Refine user requirements for preferences and expectations in online book shopping

As a user protective, these followings pages will be design with functionalities:

- i. **Sign in (New & Registered User):** Quick and secure login for users to access their saved profiles.
- ii. **Signup:** Simple registration form for new users to create an account easily.
- iii. **Reset Password:** Easy option to recover account if password is forgotten.
- iv. **Dashboard / Landing Page:** Clean homepage with overview for visitors and users.
- v. **Web Searching:** Fast search option to find books by title or author name.
- vi. **Product Category:** Books divided into School, Colleges, Universities, and other categories.
- vii. **Book Detail Page:** Clear details of price, summary, author, and stock availability.
- viii. **Add to Cart:** One click option to add selected book into cart.
- ix. **Mini Cart / Manage Items:** Easy option to update quantity or remove books from cart.
- x. **Checkout:** Simple checkout process for shipping and payment details.
- xi. **Shipment Details:** Show shipping address, courier info, and expected delivery date.
- xii. **Safe Payment:** Secure handling of card and payment information.
- xiii. **Thank You page:** Show success message with order number after purchase.
- xiv. **Order Status:** Show order confirmation and tracking ID for customer.
- xv. **Customer Support:** Easy way to contact support for any issue or complaint.
- xvi. **Responsiveness Test:** Website should work properly on mobile, tablet, and desktop.

2. Write clear and concise acceptance criteria for scenarios

Scenario 1: Finding a book

- Feature: Search
- Scenario: Searching by title

Given the subscriber is on the home page

When the subscriber types a book name in the search bar

And clicks the search button

Then a list of matching books should be displayed

And the results must show the correct book at the top

Scenario 2: Viewing book details

- Feature: Catalog
- Scenario: Checking price and stock

Given the subscriber has searched for a book

When the subscriber clicks on a specific book title

And the detail page opens up

Then the system should show the price and summary

And it must show if the book is currently in stock

Scenario 3: User Login

- Feature: Authentication
- Scenario: Successful login

Given the subscriber has a registered account

When the subscriber enters the correct email

And enters the correct password

Then the subscriber should be logged into the account

And the personal dashboard should be visible

Scenario 4: Checkout Process

- Feature: Checkout
- Scenario: Successful order placement

Given the subscriber has items in the cart

When the subscriber proceeds to checkout

And enters valid payment details

Then the order should be placed successfully

And an order confirmation should be displayed

Scenario 5: Creating a new account

- Feature: Registration
- Scenario: New user signup

Given a visitor does not have an account

When the visitor fills out the signup form

And clicks the register button

Then a new account should be created in the system

And a welcome message should be displayed

Scenario 6: Resetting password

- Feature: Account Recovery
- Scenario: Requesting a password reset

Given the subscriber has forgotten their password

When the subscriber enters their email address

And clicks the reset link button

Then an email should be sent to the subscriber

And it should contain a link to create a new password

Scenario 7: Adding items to cart

- Feature: Shopping Cart
- Scenario: Adding a book while browsing

Given the subscriber is looking at a book page

When the subscriber clicks the "Add to Cart" button

And the page stays on the current view

Then the cart icon should update the item count

And the book should be saved in the virtual cart

Scenario 8: Updating cart quantity

- Feature: Shopping Cart
- Scenario: Changing number of copies

Given the subscriber has a item in the cart

When the subscriber changes the quantity to two

And clicks the update button

Then the total price should be calculated again

And the new total must be displayed correctly

Scenario 9: Removing items from cart

- Feature: Shopping Cart
- Scenario: Deleting a book

Given there is a book in the shopping cart

When the subscriber clicks the remove icon

And confirms the deletion

Then the item should be removed from the list

And the cart total should be updated to zero

Scenario 10: Contacting support

- Feature: Help Desk
- Scenario: Sending a help message

Given the subscriber has a issue with an order

When the subscriber fills out the contact form

And clicks the submit button

Then the message should be sent to the support team

And a "Thank You" message should appear on the screen

3. Define functional requirements (authentication, catalog, search, cart, etc.)

A. Login / Accounts

- User can login with email and password.
- New user can create account.
- User can reset password if forgotten.
- User can logout from account.
- User can update profile details.

B. User Access / Roles

- Guest user can browse books and search with limited access.
- Guest user must login or register before purchase.
- Registered user can login and access account features.
- Admin has full access to manage system data.

C. Product Details & Category

- Application shows all categories, books with title, author, and price.
- User can open product details page.
- Application shows out of stock books.
- Books can be shown by category or genre.
- Featured books can be shown on homepage / landing page.

D. Search

- User can search books by title or author.
- Application shows matching results.
- If no book found, system shows message.
- User can filter books by category or price.

E. Shopping Cart

- User can add books to cart.
- User can remove books from cart.
- User can change quantity of books.
- Cart shows total price automatically.

F. Checkout / Orders

- User can enter shipping details.
- User can make payment securely.
- Application confirms order after payment.
- User can view order history.
- User can track order status.

G. Payment

- User can choose Cash on Delivery (COD).
- User can pay with debit card.
- User can pay with credit card.
- User can use online payment methods.
- Payment should be processed securely.
- System should show payment success or failure message.

4. Identify non-functional requirements (e.g., performance)

A. Performance

- Pages should load quickly.
- Search results should appear fast.
- Application should handle many users at the same time.
- Checkout process should work without delay.
- Website response should remain smooth during peak time.

B. Usability

- Website should be simple and easy to use.
- Layout should be clear for all users.
- Shopping steps should be easy to complete.
- Buttons and menus should be easy to find.
- Error messages should be clear and helpful.

C. Compatibility

- Website should work on mobile and desktop.
- Website should support major browsers.
- Website should adjust properly on different screen sizes.

D. Security

- Passwords should be protected.
- Payment details should be secure.
- Users should only access their own accounts.
- System can logout user after inactivity.
- Unauthorized login attempts should be restricted.

5. Develop user stories to describe interactions

A. Account Registration

- **Feature:** User Registration
- **Description:** This feature allows a new user to create an account for online book shopping.

Scenario: Create a new account

Given the user is on the registration page

When the user enters valid details

And clicks register

Then the account should be created successfully

And the user should be redirected to login page

B. User Login

- **Feature:** User Login
- **Description:** This feature allows registered users to login and access their account.

Scenario: Login with valid credentials

Given the user is on the login page

When the user enters valid email and password

And clicks login

Then the user should be logged in successfully

And redirected to homepage

C. Search Books

- **Feature:** Book Search
- **Description:** This feature helps users search books by title or author.

Scenario: Search a book

Given the user is on the homepage

When the user enters a book name in search bar

And clicks search

Then related books should be displayed

D. View Book Details

- **Feature:** Book Details
- **Description:** This feature shows complete information of selected book.

Scenario: View book details

Given the user is viewing search results

When the user selects a book

Then book title, author, price, and description should be displayed

E. Add to Cart

- **Feature:** Add to Cart
- **Description:** This feature allows users to add books into shopping cart.

Scenario: Add book to cart

Given the user is on book details page

When the user clicks add to cart

Then the book should be added to cart

And cart count should be updated

F. Manage Cart

- **Feature:** Update Cart
- **Description:** This feature allows users to manage items in shopping cart.

Scenario: Remove item from cart

Given the user has books in cart

When the user removes an item

Then the item should be removed successfully

And total price should be updated

G. Checkout Order

- **Feature:** Checkout
- **Description:** This feature allows users to proceed with order placement.

Scenario: Proceed to checkout

Given the user has items in cart

When the user enters shipping details

And clicks checkout

Then the user should proceed to payment page

H. Secure Payment

- **Feature:** Payment Confirmation
- **Description:** This feature confirms successful payment and order placement.

Scenario: Complete payment

Given the user enters valid payment details

When the payment is processed

Then the order should be placed successfully

And confirmation message should be displayed